

Probability & Statistics

APM_2F010_EP

Lecture Notes

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1 Lecture 1

Theorem 1: Fubini positive

Let $f : X \times Y \rightarrow [0, \infty)$, $g : X \rightarrow [0, \infty)$ and $h : Y \rightarrow [0, \infty)$ be piecewise continuous. Then:

$$\int_Y \left(\int_X f(x, y) g(x) dx \right) h(y) dy = \int_X \left(\int_Y f(x, y) h(y) dy \right) g(x) dx$$

- Works without integrability assumptions.
- Analogue: reordering double series with nonnegative terms.

Theorem 2: Fubini - Tonelli

Let $f : X \times Y \rightarrow \mathbb{R}$ (not necessarily positive), and let $g : X \rightarrow [0, \infty)$ and $h : Y \rightarrow [0, \infty)$ be piecewise continuous. If:

$$\int_Y \left(\int_X |f(x, y)| g(x) dx \right) h(y) dy < \infty$$

Then:

$$\int_Y \left(\int_X f(x, y) g(x) dx \right) h(y) dy = \int_X \left(\int_Y f(x, y) h(y) dy \right) g(x) dx$$

- Key assumption: integrable.

What is a probability?

Example 3: Rolling a fair dice

- Possible outcomes $\Omega = \{1, \dots, 6\}$.
- All outcomes have probability $1/6$.
- An event is “The result is odd”.

Proposition 4: Discrete probability space

$(\Omega, \mathbb{P})$ is a discrete prob. space. $\mathbb{P}$ is called the discrete prob. on Ω . If $\forall A \subset \Omega$, $\mathbb{P}(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}$, then $\mathbb{P}$ is called the uniform discrete probability.

Example 5: Non-discrete probability space

Let $E =$ “the extraction of a number in $[0, 1]$ uniformly at random”. E is modeled with the uniform probability $\mathbb{P}$ on the space $\Omega = [0, 1]$.

$$\forall 0 \leq a \leq b \leq 1, \quad \mathbb{P}([a, b]) = b - a$$

$(\Omega, \mathbb{P})$ cannot be a discrete prob. space.

Definition 6: Probability function

We define a prob. function $\mathbb{P} : \mathcal{P}([0, 1]) \rightarrow [0, 1]$ s.t.

1. $\mathbb{P}([0, 1]) = 1$
2. $\mathbb{P}$ is σ -additive
3. $\mathbb{P}([a, b]) = b - a \quad \forall a, b$

2 Lecture 2**2.1 General Probability Spaces****Definition 1: Sigma-algebra**

Let Ω be an arbitrary set. A σ -algebra $\mathcal{A}$ is a subset of $\mathcal{P}(\Omega)$ such that:

1. $\emptyset \in \mathcal{A}$
2. If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$
3. If $(A_n)_{n \in \mathbb{N}}$ is a sequence of events in $\mathcal{A}$, then $\bigcup_{n \in \mathbb{N}} A_n$ is also in $\mathcal{A}$.

σ -algebras are subsets of $\mathcal{P}(\Omega)$ that are closed under the basic operations on sets.

Example 2: Sigma-algebras

Let Ω be an arbitrary set and $A \subseteq \Omega$. The following are σ -algebras:

- $\mathcal{P}(\Omega)$
- $\{\emptyset, A, A^c, \Omega\}$

Definition 3: General probability space

Let Ω be an arbitrary set, $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ a sigma algebra and $\mathbb{P} : \mathcal{A} \rightarrow [0, 1]$. The triplet $(\Omega, \mathcal{A}, \mathbb{P})$ is called a (general) probability space if:

1. $\mathbb{P}[\Omega] = 1$
2. If $(A_n)_{n \in \mathbb{N}}$ is a sequence of disjoint events in $\mathcal{A}$, then $\mathbb{P}[\bigcup_{n \in \mathbb{N}} A_n] = \sum_{n \in \mathbb{N}} \mathbb{P}[A_n]$.

If conditions (i) and (ii) are verified, $\mathbb{P}$ is called a (general) probability.

Example 4: General probability space

If $(\Omega, \mathbb{P})$ is a discrete probability space, then $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$ is a general probability space.

Proposition 5: Basic properties of probability spaces

Let $\mathbb{P} : \mathcal{P}(\Omega) \rightarrow [0, 1]$ be a probability and $A, B \subseteq \Omega$. Then:

1. Let $A_1, \dots, A_n$ be a finite collection of pairwise disjoint events. Then $\mathbb{P}[A_1 \cup \dots \cup A_n] = \sum_{j=1}^n \mathbb{P}[A_j]$. This property is called finite σ -additivity or additivity.
2. If $A \subseteq B$, $\mathbb{P}[B \setminus A] = \mathbb{P}[B] - \mathbb{P}[A]$. In particular, $\mathbb{P}[A] \leq \mathbb{P}[B]$ and $\mathbb{P}[A^c] = 1 - \mathbb{P}[A]$.
3. $\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$. In particular, $\mathbb{P}[A \cup B] \leq \mathbb{P}[A] + \mathbb{P}[B]$.

2.2 The Borel σ -algebra

We need a σ -algebra which contains all intervals and is smaller than $\mathcal{P}(\Omega)$. The easiest choice is to take the smallest possible sigma algebra.

Definition 6: The Borel σ -algebra

Let $I \subseteq \mathbb{R}$ be an interval. The Borel sigma algebra $\mathcal{B}(I)$ is defined as:

$$\mathcal{B}(I) = \bigcap_{\{[a,b]: a,b \in I\} \subseteq \mathcal{A} \subseteq \mathcal{P}(I)} \mathcal{A}$$

To check that it is a good definition one has to verify that $\mathcal{B}(I)$ is indeed a sigma algebra.

Theorem 7: Uniform Probability Measure

There exists a general probability space $([0, 1], \mathcal{B}([0, 1]), \mathbb{P})$ such that $\mathbb{P}[[a, b]] = b - a$ for all $a, b \in \mathbb{R}$.

2.3 Random Variables on General Probability Spaces**Definition 8: Real random variable**

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. A function $X : \Omega \rightarrow \mathbb{R}$ is called a real random variable if and only if for any $a, b \in \mathbb{R}$ we have that:

$$X^{-1}([a, b]) := \{\omega : X(\omega) \in [a, b]\} \in \mathcal{A}$$

A real random variable X defined on a general probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is discrete if $X(\Omega)$ is at most countable.

Proposition 9: Discrete random variables as general r.v.

Let $(\Omega, \mathbb{P})$ be a discrete p.s. and $X : \Omega \rightarrow \mathbb{R}$ a r.v. Then X is a real random variable on $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$.

Remark 10: Conventions

- If $E = \mathbb{R}$, we call X a real random variable; if $E = \mathbb{R}^n$ we call X a random vector.
- “Let X be a random variable” has to be read as “Let $(\Omega, \mathbb{P})$ be a discrete probability space and X be a random variable defined on it”.
- We may denote events without brackets, and not specify the probability space. For instance: $\mathbb{P}[\{\omega : X(\omega) \in A\}] \Leftrightarrow \mathbb{P}[X \in A]$ and $\mathbb{P}[\omega] \Leftrightarrow \mathbb{P}[\{\omega\}]$.
- We may replace the intersection with a comma. For example: $\mathbb{P}[\{X \in A\} \cap \{Y \in B\}] \Leftrightarrow \mathbb{P}[X \in A, Y \in B]$.

2.4 The Law (Distribution) of a Random Variable**Proposition 11: The law of a random variable**

Let X be a real valued r.v. The law (distribution) μ_X of X is the application $\mu_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ defined by $\mu_X(C) := \mathbb{P}[X^{-1}(C)]$. In this setting, the law $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X)$ is a general probability space.

Remark 12: Two Kinds of Random Variables

In this course, a random variable $X : \Omega \rightarrow \mathbb{R}$ will be of one of two types:

- **Discrete random variables:** it exists a countable set $E \subset \mathbb{R}$ and $p_X : E \rightarrow \mathbb{R}_+$ s.t. $\forall C \in \mathcal{B}(\mathbb{R}), \mu_X(C) = \sum_{x \in C \cap E} p_X(x)$.
- **Absolutely continuous random variables:** it exists $f_X : \mathbb{R} \rightarrow \mathbb{R}_+$ s.t. $\forall a < b \in \mathbb{R}, \mu_X([a, b]) = \int_a^b f_X(u) du$.

Fundamental intuition: $\mathbb{P}(X \in [a - \epsilon, a + \epsilon]) = \mu_X([a - \epsilon, a + \epsilon]) \approx 2\epsilon f_X(a)$. Here, p_X and f_X are respectively called the discrete density and the density of the random variable X .

2.5 Conditional Probability and Independence**Definition 13: Conditional probability**

Let $(\Omega, \mathbb{P})$ be a discrete probability space and A, B two events such that $\mathbb{P}[B] > 0$. The conditional probability of A given B is denoted $\mathbb{P}[A|B]$ and defined by:

$$\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$$

Definition 14: Independence of two events

Let $(\Omega, \mathbb{P})$ be a discrete probability space and A, B two events. We say that A, B are independent if $\mathbb{P}[A \cap B] = \mathbb{P}[A]\mathbb{P}[B]$. If $\mathbb{P}[B] > 0$, we have the equivalent definition $\mathbb{P}[A|B] = \mathbb{P}[A]$.

Definition 15: Joint and marginal law

Let $X : \Omega \rightarrow E, Y : \Omega \rightarrow F$ be two r.v. The joint law $\mu_{X,Y}$ is the law of the random vector (X, Y) . In particular, for all $C_1, C_2 \in \mathcal{B}(\mathbb{R})$:

$$\mu_{X,Y}(C_1 \times C_2) = \mathbb{P}[(X, Y) \in C_1 \times C_2] = \mathbb{P}[X \in C_1, Y \in C_2]$$

In this context μ_X, μ_Y are called marginal laws. We will write $p_{X,Y}$ and $f_{X,Y}$ for respectively the discrete joint density and the joint density according to the case we study.

Proposition 16: Marginal probabilities from joint density

Let $(\Omega, \mathbb{P})$ be a discrete probability space and $X : \Omega \rightarrow E, Y : \Omega \rightarrow F$ two random variables. Then:

$$\begin{aligned} \forall y \in F, \quad p_Y(y) &= \sum_{x \in E} p_{X,Y}(x, y) \\ \forall x \in E, \quad p_X(x) &= \sum_{y \in F} p_{X,Y}(x, y) \end{aligned}$$

Definition 17: Independence of random variables

Let $(X_1, \dots, X_N)$ be real valued random variables defined all on the same p.s. $(\Omega, \mathcal{A}, \mathbb{P})$. We say that they are independent if and only if for all possible choices of $[a_1, b_1], \dots, [a_N, b_N]$ we have:

$$\mathbb{P}[X_1 \in [a_1, b_1], \dots, X_N \in [a_N, b_N]] = \prod_{i=1}^N \mathbb{P}[X_i \in [a_i, b_i]]$$

An arbitrary family $(X_i)_{i \in I}$ of random variables defined on the same probability space is independent if and only if each finite subfamily is independent.

Proposition 18: Independence under complementation

Let $(A_i)_{i \in I}$ be independent events and $I' \subseteq I$. Then the events $(B_i)_{i \in I}$ defined by

$$B_i = \begin{cases} A_i, & i \in I' \\ A_i^c, & i \in I \setminus I' \end{cases}$$

are independent.

Proposition 19: Independence and joint density

Let $X : \Omega \rightarrow E_1, Y : \Omega \rightarrow E_2$ be two random variables defined on the same discrete probability space $(\Omega, \mathbb{P})$. Then X and Y are independent if and only if:

$$\forall x \in E_1, y \in E_2, \quad p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

Proposition 20: Functions of independent random variables

Let X, Y be two independent random variables taking values in E and F respectively. Moreover, let H, K be two other sets and $f : E \rightarrow H$ and $g : F \rightarrow K$ two functions. Then the random variables $f(X)$ and $g(Y)$ are independent.

Proposition 21: The sum of independent random variables

Let X, Y be two absolutely continuous and independent random variables whose densities are f_X and f_Y . Then, the random variable $X + Y$ is absolutely continuous. Its density f_{X+Y} is given by the convolution $f_X * f_Y$ of f_X and f_Y , defined by:

$$(f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx$$

It is also true in the discrete case:

$$p_{X+Y}(z) = \sum_{x \in \mathbb{R}} p_X(x)p_Y(z-x)$$

Be careful, the notation $\sum_{x \in \mathbb{R}}$ is only licit because this is the discrete case, this is highly wrong to write it in the absolutely continuous case.